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1. Introduction

Global temperature serves as a widely used measure to represent the overall condition of the Earth's climate. Although the impacts of climate change are experienced on a local scale, climate models indicate that various global climate forcings tend to produce relatively consistent effects across the globe (Hansen, et al., 2005). Over the last century, the Earth's temperature has risen by approximately 0.9–1 ºC, whereas the Arctic region has undergone a more significant warming of 2–3 ºC during the same period (IPCC, 2018, 2021). Based on the latest forecasts, if the current rate of temperature increase persists, global warming is projected to reach 1.5 ºC by 2030 (IPCC, 2021).

Temperature as an indicator of climate change

Many studies have been conducted which indicate that climate change may have contributed to the occurrences of extreme floods, heat waves, droughts, ocean warming, coral reef bleaching, sea level rise, altered precipitation patterns, increased fire activity, more frequent and severe storms, as well as changes in species' physiology, phenology, and distribution (Rahmstorf & Coumou, 2011). For instance, during the summer of 2013, a severe heat wave struck Australia, subjecting most parts of the continent to prolonged extreme temperatures. People often speculate about whether each year is the warmest or coldest on record. To quantify global temperature changes, multiple studies have been conducted, primarily relying on data from meteorological stations (Brohan, et al., 2006; Hansen, et al., 2010; Rahmstorf & Coumou, 2011).

As observed, the effect of high temperatures range from environmental, social, to economic highlighting the importance of understanding the current state of climate affairs as a precursor to developing and adopting techniques suitable for combating any resultant effects.

Objective

The objective of the current study is to examine various aspects of climate change across the world based on temperature changes recorded between 1961 and 2020. In Particular, the current study will examine the following questions related to temperature change:

How does temperature change as an indicator of climate change vary in OECD Countries before and after Covid-19 i.e., in 2020 and after 2016 but before 2020?

A report by (OECD, 2023) shows that emissions of OECD countries reached their highest point in 2007, followed by a gradual decline until 2016, resulting in a decrease of 9%. Subsequently, emissions remained relatively stable during the period of 2017 to 2019. However, in 2020, emissions experienced a notable 7% drop due to the impact of the COVID-19 pandemic and the subsequent reduction in human activities. Despite this reduction, emissions rebounded in 2021, as reported by the International Energy Agency (IEA, 2021). By examining temperature deviations over time across the two periods, we can identify whether the rate of temperature change in OECD countries is linked to carbon emissions.

How has the average temperature changed over the years for globally?

This question helps identify long-term trends and patterns in temperature variations across different regions, enabling us to understand the implications of climate change on a global scale.

What is the overall seasonal variation of temperature around the world?

Seasonal patterns have been noted in areas such as Finland where there are different climatic zones, spanning from temperate in the southern regions to continental/subarctic in the east, and maritime/subarctic in the west and north. This leads to pronounced seasonal variations and considerable spatial diversity (Kortelainen and Karhu, 2004). This question explores the cyclic patterns in temperature for each year, helping to understand the seasonal variations and possibly identifying any shifts in the timing of seasons over time.

Are there any noticeable differences in temperature trends between developed and developing countries?

The IPCC's Fifth Assessment Report (AR5), published in 2014, extensively examined the uneven impacts of climate change on different regions and countries, with a specific focus on the disparities between developed and developing nations (IPCC 2014). According to the report, countries, particularly those in the tropics and low-lying coastal areas, are more vulnerable to the adverse effects of climate change, including rising temperatures, sea-level rise, extreme weather events, and changing precipitation patterns.

By comparing the temperature trends of Annex I (developed) and Non-Annex I (developing) countries, we can explore if there are disparities in temperature changes and assess their potential impacts on different economies.

How has the average temperature across different continents over the past two decades?

A recent climate report indicates that Europe is experiencing warming at a rate twice as fast as the global average (Serreze & Francis, 2006; Serreze, Barrett, Stroeve, & Kindig, 2009; England, Eisenman, Lutsko, & Wagner, 2021). The year 2022 witnessed extreme heat, drought, and wildfires that swept across the continent. By examining differences in average temperature change between different continents, we can establish which continent is at the highest risk of experiencing adverse climate effects due to temperature increases.

What are the temperature variations in different regions of Africa?

Between 1991 and 2021, Africa experienced a faster warming rate of approximately +0.3 ºC per decade, compared to the +0.2ºC per decade warming observed from 1961 to 1990 (World Meteorological Organization, 2022). The year 2021 ranked as either the third or fourth warmest year on record for the continent. West Africa has been identified as a climate-change hotspot, where the effects of climate change are expected to reduce crop yields and production, potentially leading to significant impacts on food security (Shepard, 2019). It is also expected that South Africa will be affected with it becoming drier, with an increase in both drought frequency and the occurrence of heat waves as we approach the end of the 21st century. Investigating temperature patterns in different African regions can help in understanding the challenges faced by this continent due to its diverse climate zones and the potential impacts on agriculture and ecosystems.

Which countries or regions have experienced the fastest temperature increase in the past 10 years?

Identifying the regions with the most rapid temperature rise can aid in prioritizing climate action and adaptation efforts for areas facing immediate challenges.

Analysis and Findings

Data Description

Data used in this study contained information related to statistics on the average surface temperature change by country, updated annually, for the period 1961 to 2019. The statistics include monthly, seasonal, and annual mean temperature anomalies, which represent the deviation of temperature from a baseline climatology ranging from 1951 to 1980. Additionally, the dataset includes the standard deviation of temperature change based on the baseline methodology. The data is sourced from the publicly available GISTEMP data, specifically the Global Surface Temperature Change data distributed by NASA's Goddard Institute for Space Studies (NASA-GISS). The data can be accessed from the following link https://www.kaggle.com/datasets/sevgisarac/temperature-change?group=bookmarked

Import Data

```
In [6]: #to work around some deprecations in trend test
        #!pip install "numpy<1.24" --user
        #!pip install pymannkendall
```

```
In [7]: import pandas as pd
        import numpy as np
        import seaborn as sns
        import plotly.express as px
        import plotly.graph_objects as go
        import matplotlib.pyplot as plt
        import statsmodels.api as sm
        from statsmodels.tsa.seasonal import seasonal_decompose
        import warnings
        warnings.filterwarnings('ignore')
```

```
In [8]: import os
        os.listdir('data')
```

```
Out[8]: ['Environment_Temperature_change_E_All_Data_NOFLAG.csv',
         'FAOSTAT_data_1-10-2022.csv',
         'FAOSTAT_data_11-24-2020.csv']
```

```
In [9]: data = pd.read_csv("data/Environment_Temperature_change_E_All_Data_NOFLAG.csv", encoding = 'ISO-8859-1')
        #data overview
        data.head()
```

Out[9]:

	Area Code	Area	Months Code	Months	Element Code	Element	Unit	Y1961	Y1962	Y1963	...	Y2010	Y2011	Y2012	Y2013	Y2014	Y2015	Y2016	Y2017	Y2018	Y2019
0	2	Afghanistan	7001	January	7271	Temperature change	°C	0.777	0.062	2.744	...	3.601	1.179	-0.583	1.233	1.755	1.943	3.416	1.201	1.996	2.951
1	2	Afghanistan	7001	January	6078	Standard Deviation	°C	1.950	1.950	1.950	...	1.950	1.950	1.950	1.950	1.950	1.950	1.950	1.950	1.950	1.950
2	2	Afghanistan	7002	February	7271	Temperature change	°C	-1.743	2.465	3.919	...	1.212	0.321	-3.201	1.494	-3.187	2.699	2.251	-0.323	2.705	0.086
3	2	Afghanistan	7002	February	6078	Standard Deviation	°C	2.597	2.597	2.597	...	2.597	2.597	2.597	2.597	2.597	2.597	2.597	2.597	2.597	2.597
4	2	Afghanistan	7003	March	7271	Temperature change	°C	0.516	1.336	0.403	...	3.390	0.748	-0.527	2.246	-0.076	-0.497	2.296	0.834	4.418	0.234

5 rows × 66 columns

```
In [10]: data2 = pd.read_csv("data/FAOSTAT_data_1-10-2022.csv", encoding = 'ISO-8859-1')
         #data overview
         data2.head()
```

Out[10]:

	ï»¿Domain Code	Domain	Area Code (FAO)	Area	Element Code	Element	Months Code	Months	Year Code	Year	Unit	Value	Flag	Flag Description
0	ET	Temperature change	2	Afghanistan	7271	Temperature change	7001	January	1961	1961	?C	0.746	Fc	Calculated data
1	ET	Temperature change	2	Afghanistan	7271	Temperature change	7001	January	1962	1962	?C	0.009	Fc	Calculated data
2	ET	Temperature change	2	Afghanistan	7271	Temperature change	7001	January	1963	1963	?C	2.695	Fc	Calculated data
3	ET	Temperature change	2	Afghanistan	7271	Temperature change	7001	January	1964	1964	?C	-5.277	Fc	Calculated data
4	ET	Temperature change	2	Afghanistan	7271	Temperature change	7001	January	1965	1965	?C	1.827	Fc	Calculated data

```
In [11]: data3 = pd.read_csv("data/FAOSTAT_data_11-24-2020.csv", encoding = 'ISO-8859-1')
         #data overview
         data3.head()
```

Out[11]:

	ï»¿"Country Code"	Country	M49 Code	ISO2 Code	ISO3 Code	Start Year	End Year
0	2	Afghanistan	4.0	AF	AFG	NaN	NaN
1	5100	Africa	2.0	NaN	X06	NaN	NaN
2	284	Ã land Islands	248.0	NaN	ALA	NaN	NaN
3	3	Albania	8.0	AL	ALB	NaN	NaN
4	4	Algeria	12.0	DZ	DZA	NaN	NaN

```
In [12]: data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 9656 entries, 0 to 9655
Data columns (total 66 columns):
 #   Column        Non-Null Count  Dtype  
---  ------        --------------  -----  
 0   Area Code     9656 non-null   int64  
 1   Area          9656 non-null   object 
 2   Months Code   9656 non-null   int64  
 3   Months        9656 non-null   object 
 4   Element Code  9656 non-null   int64  
 5   Element       9656 non-null   object 
 6   Unit          9656 non-null   object 
 7   Y1961         8287 non-null   float64
 8   Y1962         8322 non-null   float64
 9   Y1963         8294 non-null   float64
 10  Y1964         8252 non-null   float64
 11  Y1965         8281 non-null   float64
 12  Y1966         8364 non-null   float64
 13  Y1967         8347 non-null   float64
 14  Y1968         8345 non-null   float64
 15  Y1969         8326 non-null   float64
 16  Y1970         8308 non-null   float64
 17  Y1971         8303 non-null   float64
 18  Y1972         8323 non-null   float64
 19  Y1973         8394 non-null   float64
 20  Y1974         8374 non-null   float64
 21  Y1975         8280 non-null   float64
 22  Y1976         8209 non-null   float64
 23  Y1977         8257 non-null   float64
 24  Y1978         8327 non-null   float64
 25  Y1979         8290 non-null   float64
 26  Y1980         8283 non-null   float64
 27  Y1981         8276 non-null   float64
 28  Y1982         8237 non-null   float64
 29  Y1983         8205 non-null   float64
 30  Y1984         8259 non-null   float64
 31  Y1985         8216 non-null   float64
 32  Y1986         8268 non-null   float64
 33  Y1987         8284 non-null   float64
 34  Y1988         8273 non-null   float64
 35  Y1989         8257 non-null   float64
 36  Y1990         8239 non-null   float64
 37  Y1991         8158 non-null   float64
 38  Y1992         8354 non-null   float64
 39  Y1993         8315 non-null   float64
 40  Y1994         8373 non-null   float64
 41  Y1995         8409 non-null   float64
 42  Y1996         8439 non-null   float64
 43  Y1997         8309 non-null   float64
 44  Y1998         8370 non-null   float64
 45  Y1999         8324 non-null   float64
 46  Y2000         8342 non-null   float64
 47  Y2001         8241 non-null   float64
 48  Y2002         8312 non-null   float64
 49  Y2003         8390 non-null   float64
 50  Y2004         8415 non-null   float64
 51  Y2005         8424 non-null   float64
 52  Y2006         8503 non-null   float64
 53  Y2007         8534 non-null   float64
 54  Y2008         8475 non-null   float64
 55  Y2009         8419 non-null   float64
 56  Y2010         8435 non-null   float64
 57  Y2011         8437 non-null   float64
 58  Y2012         8350 non-null   float64
 59  Y2013         8427 non-null   float64
 60  Y2014         8377 non-null   float64
 61  Y2015         8361 non-null   float64
 62  Y2016         8348 non-null   float64
 63  Y2017         8366 non-null   float64
 64  Y2018         8349 non-null   float64
 65  Y2019         8365 non-null   float64
dtypes: float64(59), int64(3), object(4)
memory usage: 4.9+ MB
```

```
In [13]: data2.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 229925 entries, 0 to 229924
Data columns (total 14 columns):
 #   Column            Non-Null Count   Dtype  
---  ------            --------------   -----  
 0   ï»¿Domain Code    229925 non-null  object 
 1   Domain            229925 non-null  object 
 2   Area Code (FAO)   229925 non-null  int64  
 3   Area              229925 non-null  object 
 4   Element Code      229925 non-null  int64  
 5   Element           229925 non-null  object 
 6   Months Code       229925 non-null  int64  
 7   Months            229925 non-null  object 
 8   Year Code         229925 non-null  int64  
 9   Year              229925 non-null  int64  
 10  Unit              229925 non-null  object 
 11  Value             222012 non-null  float64
 12  Flag              229925 non-null  object 
 13  Flag Description  229925 non-null  object 
dtypes: float64(1), int64(5), object(8)
memory usage: 24.6+ MB
```

```
In [14]: data3.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 321 entries, 0 to 320
Data columns (total 7 columns):
 #   Column            Non-Null Count  Dtype  
---  ------            --------------  -----  
 0   ï»¿"Country Code"  321 non-null    int64  
 1   Country           321 non-null    object 
 2   M49 Code          304 non-null    float64
 3   ISO2 Code         245 non-null    object 
 4   ISO3 Code         257 non-null    object 
 5   Start Year        39 non-null     float64
 6   End Year          8 non-null      float64
dtypes: float64(3), int64(1), object(3)
memory usage: 17.7+ KB
```

```
In [15]: #data shape
         data.shape
```

```
Out[15]: (9656, 66)
```

```
In [16]: #number of missing entries
         #show variables with at least 1 missing entry
         data.isna().sum().sort_values(ascending = False)[data.isna().sum().sort_values(ascending = False)>0]
```

```
Out[16]: Y1991    1498
         Y1983    1451
         Y1976    1447
         Y1985    1440
         Y1982    1419
         Y1990    1417
         Y2001    1415
         Y1964    1404
         Y1989    1399
         Y1977    1399
         Y1984    1397
         Y1986    1388
         Y1988    1383
         Y1981    1380
         Y1975    1376
         Y1965    1375
         Y1980    1373
         Y1987    1372
         Y1961    1369
         Y1979    1366
         Y1963    1362
         Y1971    1353
         Y1970    1348
         Y1997    1347
         Y2002    1344
         Y1993    1341
         Y1962    1334
         Y1972    1333
         Y1999    1332
         Y1969    1330
         Y1978    1329
         Y2000    1314
         Y1968    1311
         Y1967    1309
         Y2016    1308
         Y2018    1307
         Y2012    1306
         Y1992    1302
         Y2015    1295
         Y1966    1292
         Y2019    1291
         Y2017    1290
         Y1998    1286
         Y1994    1283
         Y1974    1282
         Y2014    1279
         Y2003    1266
         Y1973    1262
         Y1995    1247
         Y2004    1241
         Y2009    1237
         Y2005    1232
         Y2013    1229
         Y2010    1221
         Y2011    1219
         Y1996    1217
         Y2008    1181
         Y2006    1153
         Y2007    1122
         dtype: int64
```

```
In [17]: #number of areas
         len(data.Area.unique())
```

```
Out[17]: 284
```

Data Preparation

Ideally, data preparation involved examining the data for missing entries and applying an appropriate treatment method. As established below, both the *Environment_Temperature_change_E_All_Data_NOFLAG* and *FAOSTAT_data_1-10-2022* datasets contain missing entries. However, since we are interested in understanding the actual temperature changes, we will exclude any rows that have missing entries to avoid introducing bias through data imputation which might lead to over or under estimation. This follows the assumption that missing entries are random i.e., missing completely at random (MCAR). This means that the missingness of the data is unrelated to the values of the variables in the dataset, both observed and unobserved.

```
In [18]: #rename variables
         data = data.rename(columns = {'Area Code':'area_code', 'Area':'area', 'Months Code':'months_code',
                                       'Months':'months','Element Code':'element_code','Element':'element','Unit':'unit'})
```

```
In [19]: data.columns
```

```
Out[19]: Index(['area_code', 'area', 'months_code', 'months', 'element_code', 'element',
                'unit', 'Y1961', 'Y1962', 'Y1963', 'Y1964', 'Y1965', 'Y1966', 'Y1967',
                'Y1968', 'Y1969', 'Y1970', 'Y1971', 'Y1972', 'Y1973', 'Y1974', 'Y1975',
                'Y1976', 'Y1977', 'Y1978', 'Y1979', 'Y1980', 'Y1981', 'Y1982', 'Y1983',
                'Y1984', 'Y1985', 'Y1986', 'Y1987', 'Y1988', 'Y1989', 'Y1990', 'Y1991',
                'Y1992', 'Y1993', 'Y1994', 'Y1995', 'Y1996', 'Y1997', 'Y1998', 'Y1999',
                'Y2000', 'Y2001', 'Y2002', 'Y2003', 'Y2004', 'Y2005', 'Y2006', 'Y2007',
                'Y2008', 'Y2009', 'Y2010', 'Y2011', 'Y2012', 'Y2013', 'Y2014', 'Y2015',
                'Y2016', 'Y2017', 'Y2018', 'Y2019'],
               dtype='object')
```

```
In [20]: #remove unnecessary attributes
         data.drop(columns=['area_code', 'months_code', 'element_code',
                   'unit'], inplace=True)
         data.head()
```

Out[20]:

	area	months	element	Y1961	Y1962	Y1963	Y1964	Y1965	Y1966	Y1967	...	Y2010	Y2011	Y2012	Y2013	Y2014	Y2015	Y2016	Y2017	Y2018	Y2019
0	Afghanistan	January	Temperature change	0.777	0.062	2.744	-5.232	1.868	3.629	-1.432	...	3.601	1.179	-0.583	1.233	1.755	1.943	3.416	1.201	1.996	2.951
1	Afghanistan	January	Standard Deviation	1.950	1.950	1.950	1.950	1.950	1.950	1.950	...	1.950	1.950	1.950	1.950	1.950	1.950	1.950	1.950	1.950	1.950
2	Afghanistan	February	Temperature change	-1.743	2.465	3.919	-0.202	-0.096	3.397	0.296	...	1.212	0.321	-3.201	1.494	-3.187	2.699	2.251	-0.323	2.705	0.086
3	Afghanistan	February	Standard Deviation	2.597	2.597	2.597	2.597	2.597	2.597	2.597	...	2.597	2.597	2.597	2.597	2.597	2.597	2.597	2.597	2.597	2.597
4	Afghanistan	March	Temperature change	0.516	1.336	0.403	1.659	-0.909	-0.069	-0.759	...	3.390	0.748	-0.527	2.246	-0.076	-0.497	2.296	0.834	4.418	0.234

5 rows × 62 columns

```
In [21]: #rename months
         months_replace = {'Dec\x96Jan\x96Feb': 'quarter_1', 'Mar\x96Apr\x96May': 'quarter_2', 'Jun\x96Jul\x96Aug': 'quarter_3',
                           'Sep\x96Oct\x96Nov': 'quarter_4'}
         data.replace(months_replace, inplace=True)
```

```
In [22]: data['months'].unique()
```

```
Out[22]: array(['January', 'February', 'March', 'April', 'May', 'June', 'July',
                'August', 'September', 'October', 'November', 'December',
                'quarter_1', 'quarter_2', 'quarter_3', 'quarter_4',
                'Meteorological year'], dtype=object)
```

```
In [23]: # Define the mapping of old values to new values (quarters)
         mapping = {
             'Dec-Jan-Feb': 'Quarter 1',
             'Mar-Apr-May': 'Quarter 2',
             'Jun-Jul-Aug': 'Quarter 3',
             'Sep-Oct-Nov': 'Quarter 4'
         }
         #map month
         data['months'] = data['months'].replace(mapping)
```

```
In [24]: data['months'].unique()
```

```
Out[24]: array(['January', 'February', 'March', 'April', 'May', 'June', 'July',
                'August', 'September', 'October', 'November', 'December',
                'quarter_1', 'quarter_2', 'quarter_3', 'quarter_4',
                'Meteorological year'], dtype=object)
```

```
In [25]: #check for missing entries in the data 2 dataframe
         data2.isna().sum()
```

```
Out[25]: ï»¿Domain Code        0
         Domain               0
         Area Code (FAO)      0
         Area                 0
         Element Code         0
         Element              0
         Months Code          0
         Months               0
         Year Code            0
         Year                 0
         Unit                 0
         Value             7913
         Flag                 0
         Flag Description     0
         dtype: int64
```

```
In [26]: data2.columns
```

```
Out[26]: Index(['ï»¿Domain Code', 'Domain', 'Area Code (FAO)', 'Area', 'Element Code',
                'Element', 'Months Code', 'Months', 'Year Code', 'Year', 'Unit',
                'Value', 'Flag', 'Flag Description'],
               dtype='object')
```

```
In [27]: #check countries given in the data
         data['area'].unique()
```

```
Out[27]: array(['Afghanistan', 'Albania', 'Algeria', 'American Samoa', 'Andorra',
                'Angola', 'Anguilla', 'Antarctica', 'Antigua and Barbuda',
                'Argentina', 'Armenia', 'Aruba', 'Australia', 'Austria',
                'Azerbaijan', 'Bahamas', 'Bahrain', 'Bangladesh', 'Barbados',
                'Belarus', 'Belgium', 'Belgium-Luxembourg', 'Belize', 'Benin',
                'Bhutan', 'Bolivia (Plurinational State of)',
                'Bosnia and Herzegovina', 'Botswana', 'Brazil',
                'British Virgin Islands', 'Brunei Darussalam', 'Bulgaria',
                'Burkina Faso', 'Burundi', 'Cabo Verde', 'Cambodia', 'Cameroon',
                'Canada', 'Cayman Islands', 'Central African Republic', 'Chad',
                'Channel Islands', 'Chile', 'China', 'China, Hong Kong SAR',
                'China, Macao SAR', 'China, mainland', 'China, Taiwan Province of',
                'Christmas Island', 'Cocos (Keeling) Islands', 'Colombia',
                'Comoros', 'Congo', 'Cook Islands', 'Costa Rica', "Côte d'Ivoire",
                'Croatia', 'Cuba', 'Cyprus', 'Czechia', 'Czechoslovakia',
                "Democratic People's Republic of Korea",
                'Democratic Republic of the Congo', 'Denmark', 'Djibouti',
                'Dominica', 'Dominican Republic', 'Ecuador', 'Egypt',
                'El Salvador', 'Equatorial Guinea', 'Eritrea', 'Estonia',
                'Eswatini', 'Ethiopia', 'Ethiopia PDR',
                'Falkland Islands (Malvinas)', 'Faroe Islands', 'Fiji', 'Finland',
                'France', 'French Guiana', 'French Polynesia',
                'French Southern and Antarctic Territories', 'Gabon', 'Gambia',
                'Georgia', 'Germany', 'Ghana', 'Gibraltar', 'Greece', 'Greenland',
                'Grenada', 'Guadeloupe', 'Guatemala', 'Guinea', 'Guinea-Bissau',
                'Guyana', 'Haiti', 'Holy See', 'Honduras', 'Hungary', 'Iceland',
                'India', 'Indonesia', 'Iran (Islamic Republic of)', 'Iraq',
                'Ireland', 'Isle of Man', 'Israel', 'Italy', 'Jamaica', 'Japan',
                'Jordan', 'Kazakhstan', 'Kenya', 'Kiribati', 'Kuwait',
                'Kyrgyzstan', "Lao People's Democratic Republic", 'Latvia',
                'Lebanon', 'Lesotho', 'Liberia', 'Libya', 'Liechtenstein',
                'Lithuania', 'Luxembourg', 'Madagascar', 'Malawi', 'Malaysia',
                'Maldives', 'Mali', 'Malta', 'Marshall Islands', 'Martinique',
                'Mauritania', 'Mauritius', 'Mayotte', 'Mexico',
                'Micronesia (Federated States of)', 'Midway Island', 'Monaco',
                'Mongolia', 'Montenegro', 'Montserrat', 'Morocco', 'Mozambique',
                'Myanmar', 'Namibia', 'Nauru', 'Nepal', 'Netherlands',
                'Netherlands Antilles (former)', 'New Caledonia', 'New Zealand',
                'Nicaragua', 'Niger', 'Nigeria', 'Niue', 'Norfolk Island',
                'North Macedonia', 'Norway', 'Oman',
                'Pacific Islands Trust Territory', 'Pakistan', 'Palau',
                'Palestine', 'Panama', 'Papua New Guinea', 'Paraguay', 'Peru',
                'Philippines', 'Pitcairn Islands', 'Poland', 'Portugal',
                'Puerto Rico', 'Qatar', 'Republic of Korea', 'Republic of Moldova',
                'Réunion', 'Romania', 'Russian Federation', 'Rwanda',
                'Saint Helena, Ascension and Tristan da Cunha',
                'Saint Kitts and Nevis', 'Saint Lucia',
                'Saint Pierre and Miquelon', 'Saint Vincent and the Grenadines',
                'Samoa', 'San Marino', 'Sao Tome and Principe', 'Saudi Arabia',
                'Senegal', 'Serbia', 'Serbia and Montenegro', 'Seychelles',
                'Sierra Leone', 'Singapore', 'Slovakia', 'Slovenia',
                'Solomon Islands', 'Somalia', 'South Africa',
                'South Georgia and the South Sandwich Islands', 'South Sudan',
                'Spain', 'Sri Lanka', 'Sudan', 'Sudan (former)', 'Suriname',
                'Svalbard and Jan Mayen Islands', 'Sweden', 'Switzerland',
                'Syrian Arab Republic', 'Tajikistan', 'Thailand', 'Timor-Leste',
                'Togo', 'Tokelau', 'Tonga', 'Trinidad and Tobago', 'Tunisia',
                'Turkey', 'Turkmenistan', 'Turks and Caicos Islands', 'Tuvalu',
                'Uganda', 'Ukraine', 'United Arab Emirates', 'United Kingdom',
                'United Republic of Tanzania', 'United States of America',
                'United States Virgin Islands', 'Uruguay', 'USSR', 'Uzbekistan',
                'Vanuatu', 'Venezuela (Bolivarian Republic of)', 'Viet Nam',
                'Wake Island', 'Wallis and Futuna Islands', 'Western Sahara',
                'Yemen', 'Yugoslav SFR', 'Zambia', 'Zimbabwe', 'World', 'Africa',
                'Eastern Africa', 'Middle Africa', 'Northern Africa',
                'Southern Africa', 'Western Africa', 'Americas',
                'Northern America', 'Central America', 'Caribbean',
                'South America', 'Asia', 'Central Asia', 'Eastern Asia',
                'Southern Asia', 'South-Eastern Asia', 'Western Asia', 'Europe',
                'Eastern Europe', 'Northern Europe', 'Southern Europe',
                'Western Europe', 'Oceania', 'Australia and New Zealand',
                'Melanesia', 'Micronesia', 'Polynesia', 'European Union',
                'Least Developed Countries', 'Land Locked Developing Countries',
                'Small Island Developing States',
                'Low Income Food Deficit Countries',
                'Net Food Importing Developing Countries', 'Annex I countries',
                'Non-Annex I countries', 'OECD'], dtype=object)
```

Drop rows with missing entries

```
In [28]: #Environment_Temperature_change_E_All_Data_NOFLAG
         data = data.dropna()
         #FAOSTAT_data_1-10-2022
         data2 = data2.dropna()
         #check for missing entries in the data 2 dataframe
         data2.isna().sum()
```

```
Out[28]: ï»¿Domain Code      0
         Domain             0
         Area Code (FAO)    0
         Area               0
         Element Code       0
         Element            0
         Months Code        0
         Months             0
         Year Code          0
         Year               0
         Unit               0
         Value              0
         Flag               0
         Flag Description   0
         dtype: int64
```

```
In [29]: #check for missing entries in the data dataframe
         data.isna().sum()
```

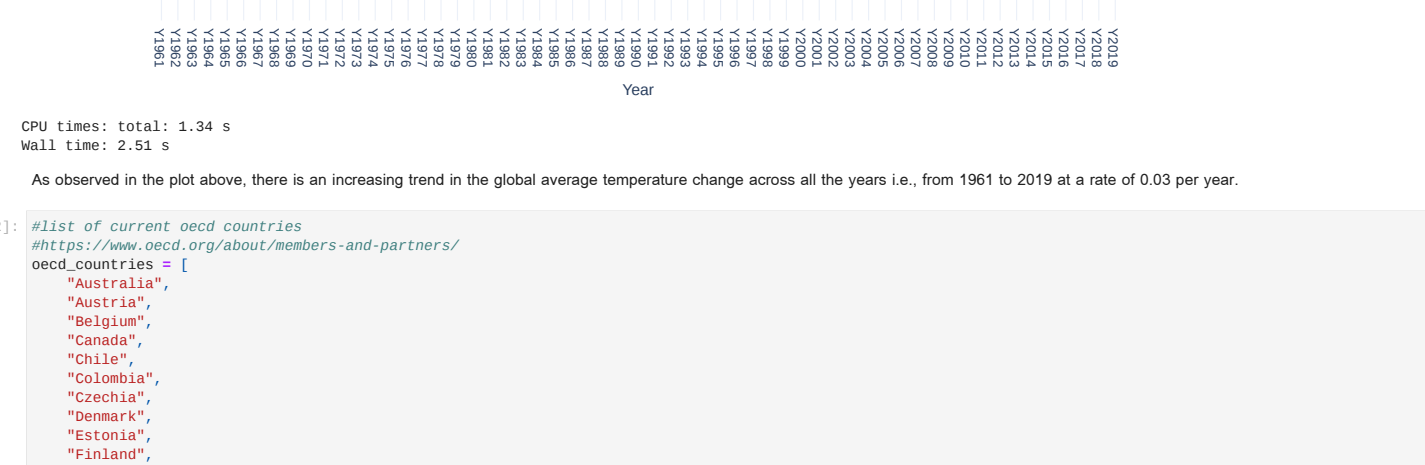
```
Out[29]: area       0
         months     0
         element    0
         Y1961      0
         Y1962      0
                   ..
         Y2015      0
         Y2016      0
         Y2017      0
         Y2018      0
         Y2019      0
         Length: 62, dtype: int64
```

Exploration

How has the average temperature changed over the years globally?

```
In [30]: #subset temperature change for the world
         world = data[(data['area'] == 'World') & (data['element'] =='Temperature change')]
         world = world[world.months.isin(['January', 'February', 'March', 'April', 'May', 'June', 'July',
                 'August', 'September', 'October', 'November', 'December'])]
```

```
In [31]: %%time
         #get world temperature averages
         temperature_averages = world.drop(['area', 'months', 'element'], axis=1).mean(axis=0)

         # Create the line chart
         fig = go.Figure()

         # Plot the temperature averages
         fig.add_trace(go.Scatter(
             x=temperature_averages.index,
             y=temperature_averages.values,
             mode='lines+markers',
             line=dict(color='blue', width=2),
             marker=dict(color='blue', size=8),
             name='Temperature Averages'
         ))

         # Fit a trend line to global temperatures
         def trendline(x, y, order=1):
             coeffs = np.polyfit(x, y, order)
             trend = np.polyval(coeffs, x)
             return trend

         # Calculate the trend line values
         trend_x = np.arange(len(temperature_averages))
         trend_y = trendline(trend_x, temperature_averages.values)

         # Get the coefficients of the trend line
         coeffs = np.polyfit(trend_x, trend_y, 1)
         slope = coeffs[0]
         intercept = coeffs[1]

         # Create the regression equation
         equation = f'y = {slope:.2f}x + {intercept:.2f}'

         # Plot the trend line
         fig.add_trace(go.Scatter(
             x=temperature_averages.index,
             y=trend_y,
             mode='lines',
             line=dict(color='black', dash='dash'),
             name='Trend Line'
         ))

         # Add the equation to the plot as annotation
         fig.add_annotation(
             xref='paper', yref='paper',
             x=0.5, y=0.9,
             text=equation,
             showarrow=False,
             font=dict(size=12),
             bgcolor='lightgrey',
             opacity=0.8
         )

         # Customize the layout
         fig.update_layout(
             title='Temperature Averages Over the Years',
             xaxis_title='Year',
             yaxis_title='Average Temperature',
             showlegend=True,
             template='plotly_white'
         )

         # Show the plot
         fig.show()
```



```
CPU times: total: 1.34 s
Wall time: 2.51 s
```

As observed in the plot above, there is an increasing trend in the global average temperature change across all the years i.e., from 1961 to 2019 at a rate of 0.03 per year.

```
In [32]: #list of current oecd countries
         #https://www.oecd.org/about/members-and-partners/
         oecd_countries = [
             "Australia",
             "Austria",
             "Belgium",
             "Canada",
             "Chile",
             "Colombia",
             "Czechia",
             "Denmark",
             "Estonia",
             "Finland",
             "France",
             "Germany",
             "Greece",
             "Hungary",
             "Iceland",
             "Ireland",
             "Israel",
             "Italy",
             "Japan",
             "Republic of Korea",
             "Latvia",
             "Lithuania",
             "Luxembourg",
             "Mexico",
             "Netherlands",
             "New Zealand",
             "Norway",
             "Poland",
             "Portugal",
             "Slovakia",
             "Slovenia",
             "Spain",
             "Sweden",
             "Switzerland",
             "Turkey",
             "United Kingdom of Great Britain and Northern Ireland",
             "United States of America"
         ]

         oecd = data2[data2.Area.isin(oecd_countries)]
```

How does temperature change as an indicator of climate change vary in OECD Countries before and after Covid-19 i.e., in 2020 and after 2016 but before 2020?

```
In [33]: #get temperature change
         oecd = oecd[oecd['Element'] =='Temperature change']
         oecd = oecd[oecd.Months.isin(['January', 'February', 'March', 'April', 'May', 'June', 'July',
                 'August', 'September', 'October', 'November', 'December'])]
```

```
In [34]: oecd_averages = oecd.groupby('Year')['Value'].mean()
```

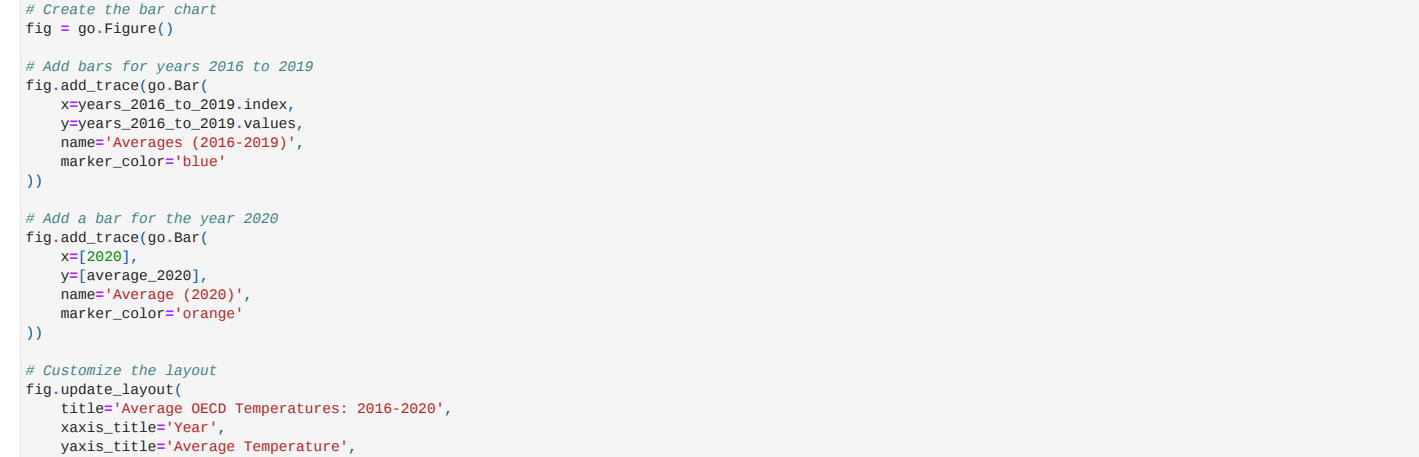
```
In [35]: len(oecd_countries)
```

```
Out[35]: 37
```

```
In [36]: len(oecd.Area.unique())
```

```
Out[36]: 37
```

```
In [37]: # Filter the data for years 2016 to 2019
         years_2016_to_2019 = oecd_averages[(oecd_averages.index >= 2016) & (oecd_averages.index <= 2019)]

         # Get the average for the year 2020
         average_2020 = oecd_averages.loc[2020]

         # Create the bar chart
         fig = go.Figure()

         # Add bars for years 2016 to 2019
         fig.add_trace(go.Bar(
             x=years_2016_to_2019.index,
             y=years_2016_to_2019.values,
             name='Averages (2016-2019)',
             marker_color='blue'
         ))

         # Add a bar for the year 2020
         fig.add_trace(go.Bar(
             x=[2020],
             y=[average_2020],
             name='Average (2020)',
             marker_color='orange'
         ))

         # Customize the layout
         fig.update_layout(
             title='Average OECD Temperatures: 2016-2020',
             xaxis_title='Year',
             yaxis_title='Average Temperature',
             showlegend=True,
             template='plotly_white'
         )

         # Show the plot
         fig.show()
```

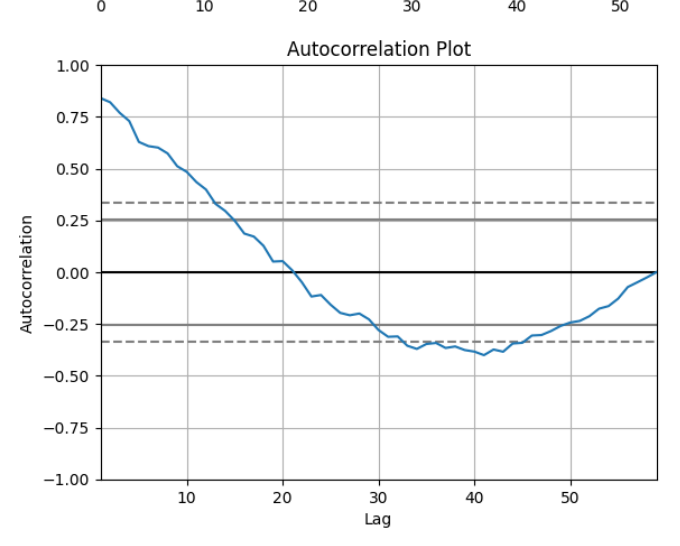


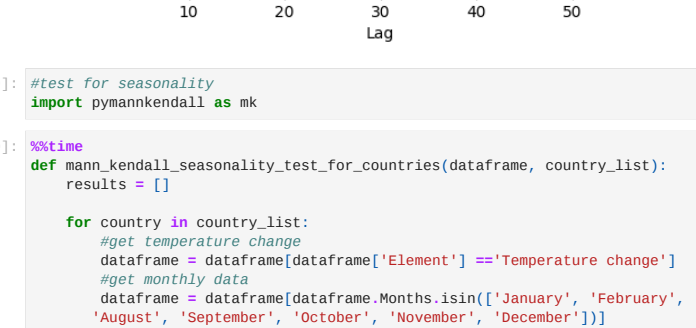
While based on the existing literature, carbon emissions in the OECD countries dropped slightly in 2020 compared to the preceeding years, we observe that temperature change increases significantly between 2016 and 2020 with a peak of approximately 2.01 units in 2020 compared to 1.54 in 2016. Theoretically, this implies that temperature change is not lagged by immediate reduction of carbon emissions.

What is the overall seasonal variation of temperature around the world?

```
In [38]: #Seasonal Decomposition
         def seasonal_decomposition(time_series):
             decomposition = seasonal_decompose(time_series['value'], model='additive', period=12)
             return decomposition

         #Autocorrelation Plot
         def plot_autocorrelation(time_series):
             pd.plotting.autocorrelation_plot(time_series['value'])
             plt.title('Autocorrelation Plot')
             plt.show()

         #get global averages
         temps = world.drop(['area', 'months', 'element'], axis = 1).mean(axis=0)
         #add to a dataframe
         world_temps = pd.DataFrame(temps, columns = ['value'])
         world_temps = world_temps.reset_index()
         world_temps.columns = ['date','value']
         #overview
         display(world_temps.head())
         # Perform seasonal decomposition
         decomposition_result = seasonal_decomposition(world_temps)

         # Plot seasonal decomposition
         decomposition_result.plot()
         plt.show()

         # Plot Autocorrelation
         plot_autocorrelation(world_temps)
```

	date	value
0	Y1961	0.134583
1	Y1962	0.061333
2	Y1963	0.130750
3	Y1964	-0.287583
4	Y1965	-0.148167





```
In [39]: #test for seasonality
         import pymannkendall as mk
```

```
In [40]: %%time
         def mann_kendall_seasonality_test_for_countries(dataframe, country_list):
             results = []

             for country in country_list:
                 #get temperature change
                 dataframe = dataframe[dataframe['Element'] =='Temperature change']
                 #get monthly data
                 dataframe = dataframe[dataframe.Months.isin(['January', 'February', 'March', 'April', 'May', 'June', 'July',
                 'August', 'September', 'October', 'November', 'December'])]
                 country_data = dataframe[dataframe['Area'] == country]['Value']


                 # Calculate the Mann-Kendall test statistic and p-value
                 res = mk.seasonal_test(country_data)

                 # Get slope
                 slope = res[6]

                 # Determine the trend direction based on the p-value
                 trend = res[0]

                 # Append the results to the list of dictionaries
                 results.append({'Country': country, 'Slope': slope, 'Trend': trend})

             # Create the DataFrame from the list of dictionaries
             results = pd.DataFrame(results)
             return results
```

```
CPU times: total: 0 ns
Wall time: 0 ns
```

```
In [41]: # Call the function
         results_dataframe = mann_kendall_seasonality_test_for_countries(data2, data2.Area.unique())
```

```
In [42]: #get the top 5 countries that show a strong increasing seasonality trend
         results_dataframe[results_dataframe['Trend'] == 'increasing'].sort_values('Slope', ascending = False).head(5)
```

Out[42]:

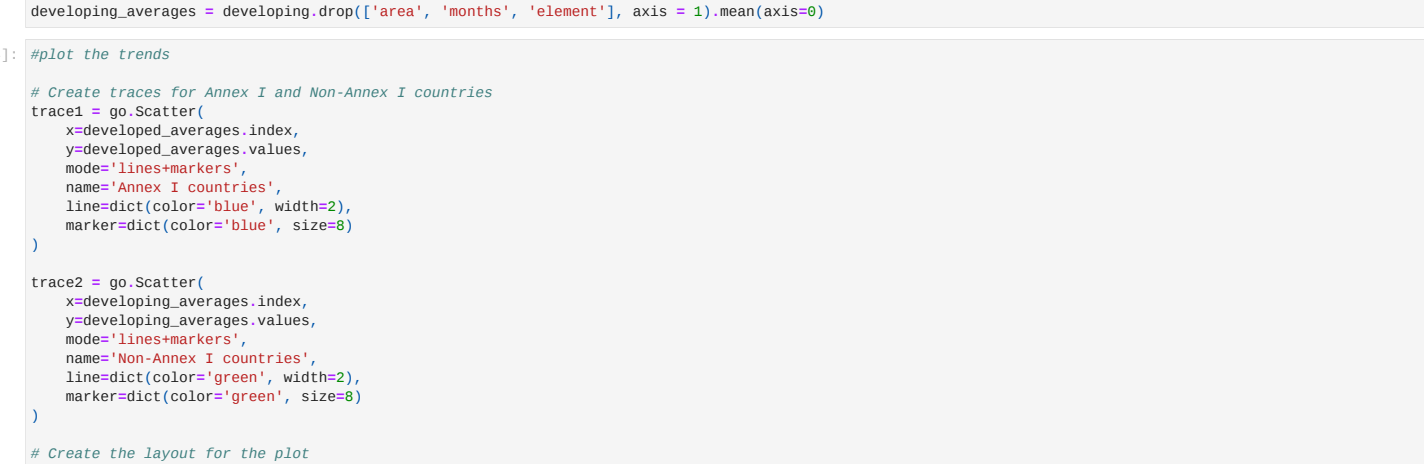
	Country	Slope	Trend
167	Palestine	295000.0	increasing
7	Antarctica	294999.0	increasing
124	Libya	294999.0	increasing
89	Gibraltar	294998.0	increasing
177	Qatar	294998.0	increasing

Examining the plot above, it is observed that overall, there is an existing seasonality in temperature change across the world. However, countries such as *Palestine, Antarctica, Libya, Gibraltar*, and *Qatar* show the strongest seasonality trend which suggests that their temperatures exhibit regular and predictable patterns across different seasons. For example, Antarctica, being a polar region, experiences extreme variations in temperature between its summer and winter months. Similarly, countries like Qatar, which have a desert climate, are known for their significant temperature fluctuations between day and night, leading to pronounced seasonality in temperature change.

Are there any noticeable differences in temperature trends between developed and developing countries?

```
In [43]: developed, developing = data[data['area']=='Annex I countries'], data[data['area'] == 'Non-Annex I countries']
         #get developing data
         developed = developed[developed['element'] =='Temperature change']
         developed = developed[developed.months.isin(['January', 'February', 'March', 'April', 'May', 'June', 'July',
                 'August', 'September', 'October', 'November', 'December'])]
         #developing
         developing = developing[developing['element'] =='Temperature change']
         developing = developing[developing.months.isin(['January', 'February', 'March', 'April', 'May', 'June', 'July',
                 'August', 'September', 'October', 'November', 'December'])]


         #get temperature averages
         developed_averages = developed.drop(['area', 'months', 'element'], axis = 1).mean(axis=0)
         #developing
         developing_averages = developing.drop(['area', 'months', 'element'], axis = 1).mean(axis=0)
```

```
In [44]: #plot the trends

         # Create traces for Annex I and Non-Annex I countries
         trace1 = go.Scatter(
             x=developed_averages.index,
             y=developed_averages.values,
             mode='lines+markers',
             name='Annex I countries',
             line=dict(color='blue', width=2),
             marker=dict(color='blue', size=8)
         )

         trace2 = go.Scatter(
             x=developing_averages.index,
             y=developing_averages.values,
             mode='lines+markers',
             name='Non-Annex I countries',
             line=dict(color='green', width=2),
             marker=dict(color='green', size=8)
         )

         # Create the layout for the plot
         layout = go.Layout(
             title='Average Temperature Change Over the Years',
             xaxis_title='Year',
             yaxis_title='Average Temperature Change',
             showlegend=True,
             template='plotly_white'
         )

         # Create the figure and add the traces to it
         fig = go.Figure(data=[trace1, trace2], layout=layout)

         # Show the plot
         fig.show()
```



As shown in the plot above, historically, temperature change among developing countries surpassed that of developed countries (Annex 1 Countries) in *1964, 1969, 1976, 1982, 1985, and 1987. After 2010, developed nations had the highest temperature changes across all years up until 2019 indicating that overall, developed nations experience a higher temperature change compared to developing nations.

How does temperature change vary between different continents

We seek to determine the continent that has the highest temperature change in the past 2 decades. The continents in question include:

```
In [45]: continents = ['Asia', 'Africa', 'Northern America', 'South America', 'Antarctica', 'Europe', 'Australia']
         continents = data[data.area.isin(continents)]

         continents.area.unique()
```

```
Out[45]: array(['Antarctica', 'Australia', 'Africa', 'Northern America',
                'South America', 'Asia', 'Europe'], dtype=object)
```

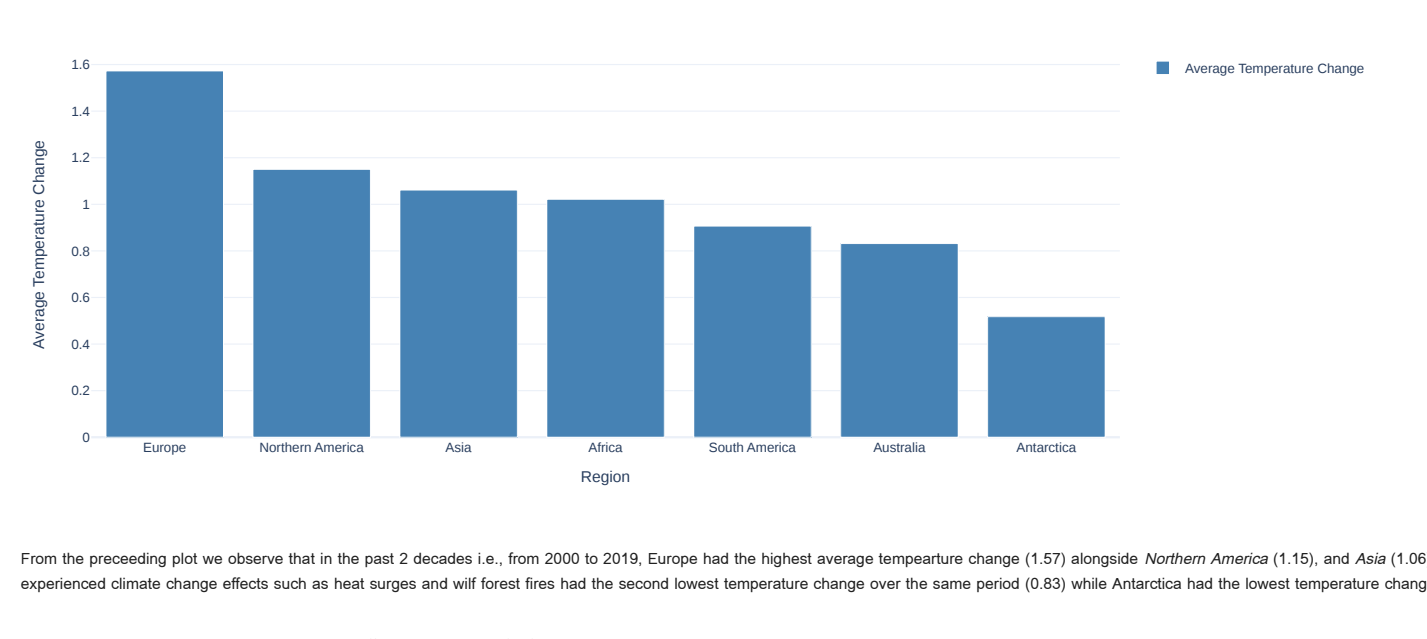
```
In [46]: #get temperature change
         continents = continents[continents['element'] =='Temperature change']
         continents = continents[continents.months.isin(['January', 'February', 'March', 'April', 'May', 'June', 'July',
                 'August', 'September', 'October', 'November', 'December'])]
         continents = continents[['area','months','element','Y2000', 'Y2001', 'Y2002', 'Y2003', 'Y2004',
                 'Y2005', 'Y2006', 'Y2007', 'Y2008', 'Y2009', 'Y2010', 'Y2011', 'Y2012',
                 'Y2013', 'Y2014', 'Y2015', 'Y2016', 'Y2017', 'Y2018', 'Y2019']]
         #data overview
         continents.head()
```

Out[46]:

	area	months	element	Y2000	Y2001	Y2002	Y2003	Y2004	Y2005	Y2006	...	Y2010	Y2011	Y2012	Y2013	Y2014	Y2015	Y2016	Y2017	Y2018	Y2019
238	Antarctica	January	Temperature change	-1.952	-0.201	1.305	0.292	0.612	1.749	0.072	...	2.022	1.525	-0.224	1.402	0.942	-0.914	-1.637	0.488	0.757	-0.509
240	Antarctica	February	Temperature change	-1.781	0.840	-0.195	0.709	1.628	0.470	-1.302	...	1.980	1.937	0.011	-1.871	-1.865	-0.679	-1.066	0.959	0.046	2.535
242	Antarctica	March	Temperature change	-1.311	-1.013	2.764	0.708	-0.253	1.989	-0.661	...	2.779	-0.917	1.480	1.626	-2.003	0.423	-0.341	1.931	1.858	1.966
244	Antarctica	April	Temperature change	-2.791	-1.726	2.972	0.245	1.556	1.247	0.934	...	-3.382	-0.984	-2.130	-0.099	1.215	0.571	0.817	-1.108	-0.026	0.183
246	Antarctica	May	Temperature change	-1.420	0.201	2.062	-1.127	-0.525	-1.167	0.450	...	2.507	-2.658	-0.475	-3.115	2.165	-1.133	1.219	1.268	3.549	0.086

5 rows × 23 columns

```
In [47]: #Compute average temperature change by region
         average_changes = continents.drop(['months', 'element'], axis = 1).groupby('area').mean().mean(axis = 1)
         #create dataframe
         average_changes = pd.DataFrame(average_changes, columns = ['Average Temperature Change'])
         average_changes = average_changes.sort_values('Average Temperature Change', ascending = False).reset_index()
         # Create a bar chart for continents
         fig = go.Figure()

         fig.add_trace(go.Bar(
             x=average_changes['area'],
             y=average_changes['Average Temperature Change'],
             marker_color='steelblue',
             name='Average Temperature Change'
         ))

         # Customize the layout
         fig.update_layout(
             title='Average Temperature Change by Region',
             xaxis_title='Region',
             yaxis_title='Average Temperature Change',
             showlegend=True,
             template='plotly_white'
         )

         # Show the plot
         fig.show()
```



From the preceeding plot we observe that in the past 2 decades i.e., from 2000 to 2019, Europe had the highest average tempearture change (1.57) alongside *Northern America* (1.15), and *Asia* (1.06). Whereas Australia despite having experienced climate change effects such as heat surges and will forest fires had the second lowest temperature change over the same period (0.83) while Antarctica had the lowest temperature change (0.52).

What are the temperature variations in different regions of Africa in the last 2 decades?

```
In [48]: #get data related to different parts of Africa
         africa = data[data.area.isin([
             'Eastern Africa', 'Middle Africa', 'Northern Africa',
             'Southern Africa', 'Western Africa'])]
         #get temperature variation
         africa = africa[africa['element'] =='Standard Deviation']
         africa = africa[africa.months.isin(['January', 'February', 'March', 'April', 'May', 'June', 'July',
                 'August', 'September', 'October', 'November', 'December'])]
         africa = africa[['area','months','element','Y2000', 'Y2001', 'Y2002', 'Y2003', 'Y2004',
                 'Y2005', 'Y2006', 'Y2007', 'Y2008', 'Y2009', 'Y2010', 'Y2011', 'Y2012',
                 'Y2013', 'Y2014', 'Y2015', 'Y2016', 'Y2017', 'Y2018', 'Y2019']]
         #data overview
         africa.head()
```

Out[48]:

	area	months	element	Y2000	Y2001	Y2002	Y2003	Y2004	Y2005	Y2006	...	Y2010	Y2011	Y2012	Y2013	Y2014	Y2015	Y2016	Y2017	Y2018	Y2019
8467	Eastern Africa	January	Standard Deviation	0.345	0.345	0.345	0.345	0.345	0.345	0.345	...	0.345	0.345	0.345	0.345	0.345	0.345	0.345	0.345	0.345	0.345
8469	Eastern Africa	February	Standard Deviation	0.338	0.338	0.338	0.338	0.338	0.338	0.338	...	0.338	0.338	0.338	0.338	0.338	0.338	0.338	0.338	0.338	0.338
8471	Eastern Africa	March	Standard Deviation	0.363	0.363	0.363	0.363	0.363	0.363	0.363	...	0.363	0.363	0.363	0.363	0.363	0.363	0.363	0.363	0.363	0.363
8473	Eastern Africa	April	Standard Deviation	0.399	0.399	0.399	0.399	0.399	0.399	0.399	...	0.399	0.399	0.399	0.399	0.399	0.399	0.399	0.399	0.399	0.399
8475	Eastern Africa	May	Standard Deviation	0.451	0.451	0.451	0.451	0.451	0.451	0.451	...	0.451	0.451	0.451	0.451	0.451	0.451	0.451	0.451	0.451	0.451

5 rows × 23 columns

```
In [49]: #Compute average standard deviation by region
         average_variations = africa.drop(['months', 'element'], axis = 1).groupby('area').mean().mean(axis = 1)
         #create dataframe
         average_variations = pd.DataFrame(average_variations, columns = ['Average Standard Deviation'])
         average_variations = average_variations.sort_values('Average Standard Deviation', ascending = False).reset_index()
         average_variations
```

Out[49]:

	area	Average Standard Deviation
0	Southern Africa	0.634500
1	Western Africa	0.552083
2	Northern Africa	0.532083
3	Eastern Africa	0.363833
4	Middle Africa	0.339250

```
In [50]: #get data related to different parts of Africa
         africa = data[data.area.isin([
             'Eastern Africa', 'Middle Africa', 'Northern Africa',
             'Southern Africa', 'Western Africa'])]
         #get temperature CHANGE
         africa = africa[africa['element'] =='Temperature change']
         africa = africa[africa.months.isin(['January', 'February', 'March', 'April', 'May', 'June', 'July',
                 'August', 'September', 'October', 'November', 'December'])]
         africa = africa[['area','months','element','Y2000', 'Y2001', 'Y2002', 'Y2003', 'Y2004',
                 'Y2005', 'Y2006', 'Y2007', 'Y2008', 'Y2009', 'Y2010', 'Y2011', 'Y2012',
                 'Y2013', 'Y2014', 'Y2015', 'Y2016', 'Y2017', 'Y2018', 'Y2019']]
         #data overview
         africa.head()
```

Out[50]:

	area	months	element	Y2000	Y2001	Y2002	Y2003	Y2004	Y2005	Y2006	...	Y2010	Y2011	Y2012	Y2013	Y2014	Y2015	Y2016	Y2017	Y2018	Y2019
8466	Eastern Africa	January	Temperature change	0.120	0.408	0.708	1.004	1.131	0.951	1.075	...	1.259	0.656	0.556	1.184	0.967	0.787	1.442	1.262	0.668	1.303
8468	Eastern Africa	February	Temperature change	0.419	0.498	0.852	1.230	0.457	1.629	1.294	...	1.274	0.777	0.838	1.198	0.699	1.375	1.443	1.163	1.178	1.245
8470	Eastern Africa	March	Temperature change	0.454	0.536	0.802	1.081	0.651	1.273	0.691	...	0.988	0.674	0.960	0.974	1.005	1.509	2.469	1.114	0.843	1.503
8472	Eastern Africa	April	Temperature change	0.662	0.675	0.586	0.891	0.348	1.110	0.482	...	1.215	1.502	0.409	0.730	0.766	0.880	1.084	1.112	0.420	2.017
8474	Eastern Africa	May	Temperature change	0.480	0.674	0.683	1.043	0.332	1.076	0.647	...	1.736	1.234	0.984	0.798	1.093	1.284	0.503	1.299	0.851	1.799

5 rows × 23 columns

```
In [51]: #Compute average temperature change by region
         average_changes = africa.drop(['months', 'element'], axis = 1).groupby('area').mean().mean(axis = 1)
         #create dataframe
         average_changes = pd.DataFrame(average_changes, columns = ['Average Temperature Change'])
         average_changes = average_changes.sort_values('Average Temperature Change', ascending = False).reset_index()
         average_changes
```

Out[51]:

	area	Average Temperature Change
0	Northern Africa	1.213729
1	Western Africa	1.096808
2	Eastern Africa	0.956954
3	Middle Africa	0.867329
4	Southern Africa	0.837617

Overall, we note that Northern Africa and Western Africa recorded the highest average temperature change over the last 2 decades . On the other hand, Middle Africa and Southern Africa had the lowest average temperature change over the same period indicating that on average, while all the regions in Africa became hotter over the past 2 decades, Northern Africa and Western Africa experienced the highest temperature change.

However, South Africa had the highest temperature variation with an average standard deviation of 0.63 compared to that of Middle Africa which had the least variaition i.e., 0.34 implying that South Africa's temperature was the least stable over the last 2 decades.

Which countries or regions have experienced the fastest temperature increase in the past 10 years?

The goal is to determine whether the observed trend in the average temperature for a given country is increasing, decreasing, or constant. To this end, we use the Mann-Kendall Trend Test to determine whether or not a trend exists in time series data. It is a non-parametric test, implying that there is no underlying assumption made about the normality of the data (Shourov & Mahmud, 2019). However, it is important to check whether the data has autocorrelation in which case a modified Mann-Kendall Trend Test is used. The following hypothesis is tested at 0.05 level of confidence:

Test for trend The hypotheses for the test are as follows:

H0 (null hypothesis): There is no trend present in the data.

HA (alternative hypothesis): A trend is present in the data. (This could be a positive or negative trend)

```
In [52]: %%time
         def mann_kendall_test_for_countries(dataframe, country_list):
             results = []

             for country in country_list:
                 #get temperature change
                 dataframe = dataframe[dataframe['Element'] =='Temperature change']
                 #get monthly data
                 dataframe = dataframe[dataframe.Months.isin(['January', 'February', 'March', 'April', 'May', 'June', 'July',
                 'August', 'September', 'October', 'November', 'December'])]
                 country_data = dataframe[dataframe['Area'] == country]['Value']


                 # Calculate the Mann-Kendall test statistic and p-value
                 res = mk.hamed_rao_modification_test(country_data)

                 # Get slope
                 slope = res[6]

                 # Determine the trend direction based on the p-value
                 trend = res[0]

                 # Append the results to the list of dictionaries
                 results.append({'Country': country, 'Slope': slope, 'Trend': trend})

             # Create the DataFrame from the list of dictionaries
             results = pd.DataFrame(results)
             return results

         # Call the function
         results_dataframe = mann_kendall_test_for_countries(data2, data2.Area.unique())
```

```
CPU times: total: 29.1 s
Wall time: 30.5 s
```

```
In [53]: #get the top 10 countries that show an increasing trend
         results_dataframe[results_dataframe['Trend'] == 'increasing'].sort_values('Slope', ascending = False).head(10)
```

Out[53]:

	Country	Slope	Trend
187	Saint Pierre and Miquelon	2.183300e+08	increasing
146	Morocco	1.326540e+08	increasing
103	India	1.107974e+08	increasing
151	Nepal	8.906908e+07	increasing
17	Bangladesh	8.226491e+07	increasing
91	Greenland	6.726546e+07	increasing
222	Tunisia	6.293155e+07	increasing
123	Liberia	6.179582e+07	increasing
189	Samoa	5.926543e+07	increasing
197	Sierra Leone	5.200290e+07	increasing

The results of the Mann-Kendall test show that the top 10 countries with the highest temperature change trends over the past 2-decade period include Saint Pierre and Miquelon, Morocco, India, Nepal, Bangladesh, Greenland, Tunisia, Liberia, Samoa, and Sierra Leone, all of which exhibited "increasing" trends in average temperature. Overall, this indicates that Saint Pierre and Miquelon, Morocco, and India had the strongest temperature change and increasing trend. Further, this implies that at the current rate, Saint Pierre and Miquelon, Morocco, and India are expected to have the highest temperature change in future.

Recommendations

Based on the findings reported in the section above, the following recommendations are made:

1. Mitigate Climate Change: Given the observed increasing trend in global average temperature change, it is crucial for all countries to accelerate efforts to mitigate climate change. This includes reducing greenhouse gas emissions, transitioning to renewable energy sources, and implementing sustainable practices across all sectors.
2. Adaptation Strategies: Countries, particularly those in Northern Africa and Western Africa, experiencing the highest average temperature changes should focus on developing robust adaptation strategies to cope with the impacts of climate change. This may involve enhancing water management, building climate-resilient infrastructure, and implementing heatwave response plans.
3. Conservation Efforts: Special attention should be given to regions like Antarctica, where temperature change as observed is relatively low, but conservation efforts are critical due to its sensitivity to climate change. Protecting fragile ecosystems and monitoring temperature changes in polar regions is essential for understanding global climate patterns.
4. Data Sharing and Monitoring: Enhance global data sharing and monitoring efforts to track temperature changes accurately. Collaborative research and sharing of climate data will provide valuable insights into regional and global climate trends, enabling better decision-making and policy formulation. Besides, this will ensure the establishment of the current global position in terms of combating climate change.
5. Climate Education and Awareness: Promote climate education and awareness among the public, policymakers, and businesses. Understanding the implications of climate change and the importance of collective action can lead to greater support for sustainable practices and climate policies.
6. International Cooperation: Encourage international cooperation and partnerships to address climate change on a global scale. Collaborative efforts among countries, organizations, and stakeholders are vital to effectively combat climate change and achieve common climate goals such as the Kyoto Protocol to ensure the mitigation of climate change and its corresponding effects.

Conclusion

The primary objective of the current study was to examine temperature change and its variation as measured by temperature standard deviation over time using data provided by FAOSTAT. To this end, the analysis of temperature change data as conducted in the current work shows that there exists a clear increasing trend in global average temperature over the past six decades. This trend, driven by human-induced climate change, poses significant challenges to ecosystems, economies, and societies worldwide. Countries in Northern Africa and Western Africa have experienced the highest average temperature changes, necessitating urgent adaptation measures. Moreover, the observed seasonality in temperature change highlights the need for tailored climate strategies for countries like Palestine, Antarctica, Libya, Gibraltar, and Qatar.

While developed countries have historically contributed more to carbon emissions, the recent trend indicates that temperature change has increased significantly in these nations as well. This finding emphasizes the importance of continued efforts to reduce emissions and combat climate change on a global scale.

Overall, the results empasize the urgency of taking action to limit global temperature rise as an indicator of climate change and address the impacts of climate change. Implementing the recommended measures and fostering international cooperation are essential steps toward a sustainable and climate-resilient future.

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0for%20Africa.

```
In [54]: #!pip install ipython --user
```

```
In [55]: import io
         import os
         from nbformat import current

         count_words_markdown = 0
         count_words_heading = 0
         count_words_code = 0

         for root, dirs, files in os.walk("."):
             for file in files:
                 if file.endswith("Data Exploration.ipynb") and not file.endswith("checkpoint.ipynb") :
                     #print(os.path.join(root, file))
                     with io.open(os.path.join(root, file), 'r', encoding='utf-8') as f:
                         nb = current.read(f, 'json')

                     word_count_markdown = 0
                     word_count_heading = 0
                     word_count_code = 0
                     for cell in nb.worksheets[0].cells:
                         if cell.cell_type == "markdown":
                             word_count_markdown += len(cell['source'].replace('#', '').lstrip().split(' '))
                         elif cell.cell_type == "heading":
                             word_count_heading += len(cell['source'].replace('#', '').lstrip().split(' '))
                         elif cell.cell_type == "code":
                             word_count_code += len(cell['input'].replace('#', '').lstrip().split(' '))
                     count_words_markdown += word_count_markdown
                     count_words_heading += word_count_heading
                     count_words_code += word_count_code

         print("There are {} Words in the reports' markdown" .format(count_words_markdown))
         print("Total word count in the report excluding the codes is {} " .format(count_words_markdown + count_words_heading))
```

```
There are 3058 Words in the reports' markdown
Total word count in the report excluding the codes is 3063
```